

CLIFFGUARD: Research Proposal

An Edge-Native, Quantization-Aware, Black-Box-Tolerant, RL-Adapted Defense System

Against Prompt Injection at the Safety Cliff

Full framework blueprint, Phase A/B empirical results, and complete evaluation roadmap

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Abstract

CLIFFGUARD is a stateless, online-RL-adapted, defense-in-depth system for prompt injection on edge-deployed large language models (LLMs) operating across heterogeneous post-training quantization (PTQ) regimes. It is motivated by a body of evidence that PTQ degrades RLHF-installed safety behavior non-linearly [??] and that the refusal direction in the residual stream can be probed as a sensor [??].

This document is the full research proposal for CLIFFGUARD. It describes: (1) the complete system architecture and its eight named components; (2) the formal theoretical foundations (decoupling theorem, CUSUM, LinUCB formulation); (3) the threat model of nine adversaries; (4) the five pre-registered hypotheses and the five-fold evaluation protocol; (5) a precise account of what has been **implemented**, what has been **tested with real data**, and what **remains planned**; (6) first empirical results from Folds A and B on Google Colab T4 (Llama-3.2-3B-Instruct, FP16 and NF4); (7) comparison to eleven existing defense systems; and (8) timeline, resources, and responsible-disclosure plan.

Current empirical status (May 2026): Fold A calibration and Fold B cliff measurement are complete on one model family and two quantization schemes. Geometric cliff metric $\Delta_{\text{cliff}} = 0.167$ (0.67κ , where $\kappa = 0.25$); Wasserstein metric $\Delta_{W\text{-cliff}} = 0.014$; behavioral proxy $\Delta_{B\text{-cliff}} = 0.000$. Hypothesis H1 (cliff existence) is not accepted for this configuration. The predicted cliff zone (GGUF Q3_K.M and below) has not yet been evaluated. Folds C–E and Hypotheses H2–H5 are planned.

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1 Motivation and Problem Statement

1.1 The Safety Cliff

Modern instruction-tuned LLMs encode safety alignment via RLHF [??]. ? showed this alignment is mediated by a single linear direction $\hat{r}$ in the residual stream; ablating $\hat{r}$ raises attack success rates dramatically. ? (NeurIPS 2025) extended this to show that *harmfulness* and *refusal* are encoded in **separate** directions: $\hat{h}$ at the user-instruction token, $\hat{r}$ at the post-instruction token.

Post-training quantization (PTQ)—reducing weights from FP16 to INT8, NF4, AWQ-INT4, GGUF Q4/Q3/Q2—enables edge deployment but may destroy this safety subspace:

- ? showed GPTQ-3-bit on Llama-2-13B-Chat drops toxicity safety by ~ 50 points while MMLU falls only ~ 8 points.
- ? (NeurIPS 2024) demonstrated the *cliff-exploit attack*: adversaries craft FP16 model weights that appear safe, where malicious behavior activates *only after* GGUF quantization, with ASR deltas of $\Delta = 88.7\%$ (insecure code), $\Delta = 85.0\%$ (content injection), and $\Delta = 30.1\%$ (refusal bypass).
- ? (ICLR 2025) showed GGUF k-quants’ MSE objective does not preserve refusal-related activation geometry.
- ? showed PTQ optimized for perplexity reconstruction systematically degrades behavioral alignment even when perplexity is preserved.

We define the **safety cliff** as the quantization level at which refusal degradation crosses a pre-registered threshold $\kappa \geq 0.25$. Hypothesis H1 pre-registers that this boundary lies at the Q4_K_M $\rightarrow$ Q3_K_M transition for at least 2 of 3 model families.

1.2 Why Existing Defenses Are Insufficient

No existing prompt-injection defense system was designed for this failure mode in an edge-hardware envelope. Llama Guard [?], PromptGuard 2, NeMo Guardrails, Rebuff, SecAlign, Constitutional Classifiers [?], LlamaFirewall [?], CaMeL, AEGIS [?], and SmoothLLM are all designed for server-side or full-precision deployment, are not quantization-aware, and provide no formal portability guarantee across schemes. CLIFFGUARD is, to our knowledge, the only system with all of: quantization-aware calibration, FPR-decoupling theorem, online bandit adaptation, four-tier MVDS, weight-hash attestation, and a black-box fallback path.

1.3 Novel Adversary: A7 Quantization-Cliff Exploiter

We introduce **A7**: an adversary who crafts natural-language prompts whose refusal margin is just above τ_q under FP16 but falls below it under Q3_K_M / NF4. Because the FP16 model refuses, such prompts pass content moderation at upload time; they succeed silently after quantization for edge deployment. This adversary is enabled by the Egashira, Hong, and Mind the Gap results and has no published defense.

2 System Architecture: Eight Components

CLIFFGUARD comprises eight named components operating in a coordinated pipeline.

2.1 VESTIBULE — Stateless Write-Side Input Gates

VESTIBULE is a stack of fast, stateless gates over the raw prompt and any retrieved/tool content:

VESTIBULE-LZ LZ4/zlib compression ratio $\rho_{LZ}(x) = |z(x)|/|x|$. GCG adversarial suffixes are less compressible than English (near-random tokens); ArtPrompt [?] is more compressible. Two-sided gate; $\approx 20 \mu\text{s}/\text{KB}$, effectively free.

VESTIBULE-PS Provenance-aware spotlighting [?]. Datamarks untrusted spans with a session-secret marker; asks the model to track provenance. Cuts injection ASR from $> 50\%$ to $< 2\%$ on capable models. On 2 GB tiers: hard-delimited prompts only (datamarking omitted).

VESTIBULE-ID Instruction-density gate: verb–entity collision counting to detect instruction hijacking without perplexity.

VESTIBULE-EA Encoding-anomaly gate: non-printable character density, base-N detection, ASCII art density (targets A6).

VESTIBULE is stateless across requests; its configuration is identical across all hardware tiers.

Implemented as Scaffolding (Phase A)

VESTIBULE is implemented as Phase A scaffolding. All four sub-gates produce verdicts on synthetic data. 939 tests pass. No real-traffic calibration has been run.

2.2 PROBE — Quantization-Aware White-Box Observer

PROBE attaches to the protected model’s forward pass via registration hooks (`register_forward_hook` in transformers; `cb_eval` callback in llama.cpp). It does not introduce a separate classifier.

PROBE-RM (Refusal Margin)

Pre-compute the refusal direction $\hat{r}_{M,q}$ via Arditi difference-in-means [?] at layer ℓ on a calibration set $\mathcal{C}$. At inference, compute the **refusal margin**:

$$\rho(x) = \frac{\langle h_\ell(x), \hat{r} \rangle}{\|\hat{r}\|} \quad (1)$$

at the post-instruction token $t_{\text{post-inst}}$, before generation. Gate fires when $\rho(x) < \tau_q$, where τ_q is a per- (M, q) quantile of the benign calibration distribution at $\text{FPR} = \alpha = 0.05$.

PROBE-MT (Margin Trajectory)

Computes ρ_ℓ at a small set of layers (e.g., 25%/50%/75% depth). Defines $\dot{\rho} = \rho_{\text{late}} - \rho_{\text{mid}}$. Adversarial suffixes that redirect the model often show characteristic non-monotone margin trajectories absent in benign prompts.

PROBE-HD (Harmfulness Direction)

Following ?, projects the residual stream at the *user-instruction* token t_{inst} onto the harmfulness direction $\hat{h}$. Together with PROBE-RM this forms a 2-D safety subspace probe: ρ_r catches response-stage refusal, ρ_h catches request-stage harmfulness recognition.

Quantization-Aware Calibration

The calibration table $(\hat{r}_{M,q}, \tau_q, \mu_0^{(q)})$ is recomputed once per (M, q) pair on a fixed calibration set $\mathcal{C}$. This is the operational content of the FPR-decoupling theorem (§3.2): only the calibration table is quantization-specific; the gate code is portable.

Tested / Empirically Validated

PROBE-RM is tested with real model inference. Run: **Llama-3.2-3B-Instruct**, FP16 and NF4, layer 14, Colab T4. Calibrated thresholds: $\tau_{q\text{FP16}} = 0.09742$, $\tau_{q\text{NF4}} = 0.09827$ ($\Delta\tau_q = +0.00085$). 400 benign + 200 harmful prompts. Fold A complete.

Implemented as Scaffolding (Phase A)

PROBE-MT and PROBE-HD are implemented as Phase A scaffolding. They accept synthetic hidden-state arrays and produce correct output shapes. Not tested with real inference.

2.3 B-PROBE — Black-Box Fallback Observer

For closed-weight API endpoints and NPU-frozen graphs without residual-stream access.

B-PROBE-LOGIT

Let $\ell(q) = (\log p(v_1 | q), \dots, \log p(v_k | q))$ be the top- k log-probability vector at the first response token. Train a logistic head $\varphi_\theta : \mathbb{R}^k \rightarrow \mathbb{R}$ on the calibration set $\mathcal{C}$ with labels (refused/complied). B-PROBE-LOGIT margin: $m_{\text{BL}}(q) = \varphi_\theta(\ell(q))$.

The first-token logit gap between {"I cannot", "I'm sorry"} and {"Sure", "Here"} is a noisy linear functional of $\hat{r}^\top z$ via the unembedding W_U . FPR portability holds (Corollary 2); TPR is strictly weaker and pre-registered as Hypothesis H3.

B-PROBE-CONSISTENCY

Generate $N \in \{2, 3\}$ paraphrases of the prompt using a *local* paraphraser (1B distilled, different family from test model). Query the closed endpoint. Compute pairwise Jensen–Shannon divergence of first-token distributions. High divergence indicates the prompt is on the cliff; small perturbations flip behavior.

Implemented as Scaffolding (Phase A)

Both B-PROBE sub-primitives are Phase A scaffolding. Top-k logprob integration with a real API or inference engine has not been tested.

2.4 TRIPWIRE — Streaming Change-Point and Entropy Monitor

TRIPWIRE operates *during decoding* without buffering outputs.

TRIPWIRE-H (Per-Token Entropy CUSUM)

Per-token entropy $H_t = -\sum_v p_t(v) \log p_t(v)$ over the model’s top- k softmax. One-sided CUSUM on entropy residuals:

$$S_t = \max(0, S_{t-1} + (H_t - \mu_0 - \delta/2)), \quad \text{alarm if } S_t > h. \quad (2)$$

Threshold h tuned for $\text{ARL}_0 \approx 1000$ benign tokens (roughly one false alarm per 50–100 sessions at typical conversation lengths). μ_0 is the per- (M, q) benign mean estimated on the calibration fold.

An injection pushing the model toward a forbidden completion typically shows an entropy depression (high-confidence forbidden token) followed by an entropy spike (model “starts and stalls”). GCG-perturbed contexts often show abnormal early entropy; many-shot injections show monotonic NLL decrease detectable by the CUSUM.

TRIPWIRE-R (KenLM Reference-Ratio Gate)

Train a KenLM 5-gram model offline on a benign instruction corpus. At decode time: $\Delta_t = \log P_M(y_t \mid \text{ctx}) - \log P_{\text{ref}}(y_t \mid \text{ctx})$. This approximates the Neyman–Pearson log-likelihood ratio. Combined with TRIPWIRE-H via CUSUM over $\alpha H_t + \beta \Delta_t$ where (α, β) is fit on calibration traces. Because the reference n -gram is fixed, quantization shifts only the $\log P_M$ term; the ratio absorbs scheme-specific variation better than bare perplexity.

Implemented as Scaffolding (Phase A)

Both TRIPWIRE sub-primitives are Phase A scaffolding. Streaming integration with real model decoding has not been tested. KenLM is not yet trained on any corpus.

2.5 CONDUCTOR — LinUCB Contextual-Bandit Gate Orchestrator

CONDUCTOR selects a gate-weight vector per request. It is the only learnable component; all others are calibration-only.

Formulation

Arms. A finite menu $\mathcal{A}$ of preset gate-weight vectors, $|\mathcal{A}| = 8\text{--}16$.

Context. $x_t \in \mathbb{R}^d$ with $d \approx 32$: prompt length, character entropy, language ID, has-tool-context flag, hardware tier, recent-incident EWMA, TRIPWIRE-H baseline drift estimate. **No payload content is stored.**

Policy. LinUCB [?]:

$$a_t = \arg \max_a \left(x_t^\top \hat{\theta}_a + \alpha \sqrt{x_t^\top A_a^{-1} x_t} \right), \quad (3)$$

with $A_a = \lambda I + \sum_{s:a_s=a} x_s x_s^\top$, $b_a = \sum r_s x_s$, $\hat{\theta}_a = A_a^{-1} b_a$. Regret: $\tilde{O}(\sqrt{dT})$ in the stochastic regime.

Adversarial fallback. EXP3.S [?] when incident-rate EWMA crosses a threshold, giving $\tilde{O}(\sqrt{KTS})$ regret for S distribution switches.

Pre-registered reward function.

$$r_t = \mathbb{I}[\text{served} \wedge \text{clean}] - \mathbb{I}[\text{served} \wedge \text{injected}] - 0.2 \cdot \mathbb{I}[\text{blocked} \wedge \text{benign}]. \quad (4)$$

Updates on: LOOKOUT verdict, canary-token leakage, tool-call audit failure, user complaint flag.

Cold-start. Fresh deployments use the static maximum-coverage policy until $O(d)$ requests per arm accrue.

Safe rollback (A8 defense). If the cumulative ABR over W rounds exceeds a pre-registered $A_{\max}$ for two consecutive windows, revert to the static maximum-coverage policy for K rounds, then resume exploration.

Implemented as Scaffolding (Phase A)

CONDUCTOR is Phase A scaffolding. LinUCB and EXP3.S are implemented and accept synthetic reward signals. No live deployment or bandit training has been run. Fold D (bandit drift) is planned.

2.6 LOOKOUT — Output-Side Judge and Canary Loop

LOOKOUT-CT (Canary Token Loop)

Per-session CSPRNG-generated canary token sequence inserted into the system prompt. LOOKOUT scans output streams (via a Count-Min sketch keyed by the session secret) for any echo. Canary is per-session, never logged, discarded at session end. Bloom filter: $m = 256$ bits, $k = 3$ hashes; FPR ≈ 0.001 at typical canary lengths. Total session state: ≈ 32 bytes.

LOOKOUT-JG (KL-Drift / Mutation Judge)

Generate $N \in \{2, 3\}$ character-level mutations of the input. Take only the first-token distributions of the model on each. Compute pairwise JS divergence. High divergence on the *first token* indicates an unstable input—characteristic of GCG/AutoDAN-perturbed prompts. Cost: N extra prefill calls of one token each (not full responses). On 8 GB tier: $N = 3$; Pi 5: $N = 2$; 2 GB tier: omitted.

LOOKOUT Classifier

On 8 GB tier: Llama-Guard-3-1B-INT4 [?] (440 MB) or DeBERTa-86M. On Pi 5: DeBERTa-22M or PromptGuard-2-22M [?]. On 2 GB tier: rules only. Provides the reward signal r_t to CONDUCTOR.

Implemented as Scaffolding (Phase A)

All LOOKOUT sub-components are Phase A scaffolding. Canary logic, JG, and classifier wrappers accept synthetic data. No real-inference integration tested.

2.7 LADDER — Four-Tier MVDS Hardware Router

LADDER is configuration, not learning. Selects which gates run at what depth based on detected hardware and observability.

Table 1: Four hardware tiers and their gate configurations.

Tier	Hardware	Schemes	Active Gates	Scope
A	RTX 5060 8 GB	FP16, NF4, AWQ-INT4, Q4-Q6	All 12 gates (incl. LOOKOUT-JG)	Full stack, LinUCB $ \mathcal{A} = 16$
B	Raspberry Pi 5 8 GB CPU	GGUF Q4_K_M, Q3_K_M	11 gates (LOOKOUT-JG $N = 2$)	LinUCB $ \mathcal{A} = 8$; B-PROBE-CONSISTENCY
C	2 GB embedded (RK3588, Jetson Orin, Pi 4)	Q3_K_M, Q2_K, RKNN W8A8	VESTIBULE-LZ, PROBE-RM (1 layer), Page-Hinkley, LOOKOUT-CT	No bandit ; static weights; NOT defended
C+	2 GB + PromptGuard-2-22M-INT4	Q3_K_M, RKNN W8A8	VESTIBULE-LZ/PS, B-PROBE-LOGIT, PROBE-RM, LOOKOUT-CT, PG2	Static weights; H5 tests this tier

Plain-language statement on Tier C: Tier C runs Q3_K_M models in 2 GB—the very regime where Egashira and Hong et al. cliffs are most likely. The minimal gate stack raises attacker cost only marginally. **Tier C is NOT FOR OPEN-DOMAIN ADVERSARIAL USE.** Suitable only for single-task Jatmo-style [?] assistants with a fixed input grammar and no agentic tool use.

Implemented as Scaffolding (Phase A)

LADDER is implemented as a configuration-based tier dispatcher. All tier routing logic is scaffolded and tested with synthetic data.

2.8 ATTEST — Boot-Time Weight-Hash Attestation

Boot-time SHA-256 of the GGUF/safetensors blob compared against a vendor-signed manifest. Defends against A2 (Egashira-style poisoned weights [?]).

Llama, Qwen, Mistral, Gemma publish per-file SHA-256 on Hugging Face. For unsigned community quantizations, ATTEST falls back to first-use trust with a sticky local hash.

$$\text{ATTEST}(M) = \begin{cases} \text{ALLOW} & \text{if SHA256(blob) = manifest hash} \\ \text{BLOCK} & \text{if mismatch} \end{cases} \quad (5)$$

Implemented as Scaffolding (Phase A)

ATTEST-WH is Phase A scaffolding. The SHA-256 hashing and manifest comparison logic are implemented. No live boot-time integration with an inference engine has been tested.

3 Theoretical Foundations

3.1 Geometric Assumption

Definition 1 (Safety Subspace, G1). *Let $z_t(q)$ denote the residual-stream activation at token t for prompt q . There exists a subspace $S = \text{span}\{\hat{r}, \hat{h}\} \subset \mathbb{R}^d$ such that for any distribution P over prompts,*

$$\|\Pi_S z_{t_{\text{inst}}}(q) - \Pi_S z_{t_{\text{inst}}}(q')\| \geq \xi \cdot \|z_{t_{\text{inst}}}(q) - z_{t_{\text{inst}}}(q')\|$$

in expectation over $(q, q') \in \mathcal{H} \times \mathcal{B}$, for some $\xi > 0$.

G1 is empirically supported by ? and ? for instruction-tuned models at 7B–72B parameters.

Sensitivity bound. If quantization $\tilde{q}$ acts as a Lipschitz operator on the residual stream with constant $L_{\tilde{q}}$, then

$$\|\pi(q) - \pi(q')\| \leq L_{\tilde{q}} \cdot (\|\delta_z\|_2 + 2\varepsilon_{\tilde{q}}), \quad (6)$$

where $\varepsilon_{\tilde{q}}$ is the per-quantization residual reconstruction error and δ_z is the induced activation perturbation. The **cliff is the regime where $\varepsilon_{\tilde{q}}$ inflates enough that $\pi(q)$ flips sign on harmful prompts.**

3.2 FPR-Decoupling Theorem

Theorem 1 (FPR Decoupling, White-Box). *Let $g(x) = \mathbb{I}[\phi(x; M_q) > \tau_q]$ be a write-side gate operating on (a) input strings and/or (b) summary statistics of model outputs (top- k log-probabilities, refusal-margin scalars). Let τ_q be the empirical $(1 - \alpha)$ -quantile of $\phi(\cdot; M_q)$ on a calibration set $\mathcal{C} \sim \mathcal{B}$. Then*

$$|\text{FPR}_{g,q} - \text{FPR}_{g,q^*}| \leq \text{KS}(F_q, F_{q^*}) \leq d_{q,q'}$$

as $|\mathcal{C}| \rightarrow \infty$, where $d_{q,q'}$ is the Kolmogorov–Smirnov distance between benign-margin distributions.

Proof sketch. By Glivenko–Cantelli, the empirical quantile of $\phi(\cdot; M_q)$ on $\mathcal{C}$ converges uniformly to the population quantile at rate $O(|\mathcal{C}|^{-1/2})$, regardless of the distribution of ϕ . Thus the realized FPR of each gate is $\alpha + o(1)$ at any fixed α . Composition by AND/OR/weighted-sum is a finite function of finitely many α -controlled gates; the result follows. The KS bound follows from the definition of CDF inverse and KS distance. $\square$

Corollary 2 (FPR Decoupling, Black-Box). *The same theorem applies to B-PROBE-LOGIT’s logistic head $\varphi_\theta : \mathbb{R}^k \rightarrow \mathbb{R}$ trained on (prompt, refused/complied) pairs. TPR is not decoupled in either case.*

Honest scope. Theorem 1 protects FPR only. It says nothing about TPR. The cliff is *defined* as the regime where TPR collapses while FPR is maintained.

Sensitivity corollary. With $|\mathcal{C}| < 100$, the KS estimation error dominates, giving an $O(1/\sqrt{|\mathcal{C}|})$ inflation of the bound. Deployments with $|\mathcal{C}| = 2000$ from Anthropic-HH suffice; deployments with $|\mathcal{C}| < 100$ must accept a wider FPR portability band.

3.3 Information-Theoretic Framing

TRIPWIRE approximates the Neyman–Pearson log-likelihood ratio $\Lambda = \log P_1/P_0$, which CUSUM (Page 1954) accumulates with optimal expected detection delay (Lorden 1971, Mous-

takides 1986). Pure perplexity gates approximate $\log P_0$ alone; TRIPWIRE approximates the ratio by combining a fixed KenLM reference with the model’s own per-step entropy.

3.4 RL-Theoretic Framing

CONDUCTOR is a contextual adversarial bandit. LinUCB gives $\tilde{O}(\sqrt{dT})$ regret in the stochastic regime. Under non-stationary attack distributions, EXP3.S [?] gives $\tilde{O}(\sqrt{KTS})$ for S switches—appropriate during coordinated attack campaigns.

4 Threat Model (A1–A9)

We adopt a Greshake-style [?] hierarchy expanded for the edge-quantized setting. No persistent storage of user payloads; only ephemeral session windows, RL policy weights, sketch counters, and signal buffers may be retained.

Table 2: Threat model: nine adversaries, capabilities, and primary CLIFFGUARD defenses. A7 is the novel adversary introduced by this work.

ID	Adversary	Capability	Primary Defense
A1	Direct injector	NL hijacks, DAN, role-play, persuasion	VESTIBULE, LOOKOUT
A2	Indirect injector / poisoned-weight attacker [?]	Embeds instructions in RAG; FP16-benign / Q3_K_M-malicious weights	VESTIBULE-PS, ATTEST + LADDER
A3	Optimizer [?]	GCG, AutoDAN, AmpleGCG adversarial suffixes	PROBE, VESTIBULE-LZ
A4	Iterator	PAIR, TAP, Crescendo [?], black-box query loops	TRIPWIRE, LOOKOUT
A5	Scaler	Best-of-N [?], many-shot, sampling variance	TRIPWIRE-H, LOOKOUT-CT
A6	Encoder	ArtPrompt [?], cipher, low-resource language	TRIPWIRE-R
A7	Quantization-cliff exploiter	Natural-language prompts that refuse at FP16 but comply at Q3_K_M / NF4	LADDER + ATTEST + cross-tier rollback
A8	Defender-aware adversary	Knows policy weights and calibration tables (Kerckhoffs assumption)	CONDUCTOR safe-rollback
A9	Closed-weight black-box endpoint adversary	Targets API endpoint; only text and top- k logprobs observable	B-PROBE + LOOKOUT-JG

5 Cliff Metrics: Formal Definitions

5.1 Geometric Cliff Δ_{cliff}

Let M_q be the model quantized to scheme q and $\mathcal{H}$ a held-out harmful set. Define the **separation** at scheme q :

$$\Delta_q = \text{med}(\mathcal{R}_q^{\mathcal{B}}) - \text{med}(\mathcal{R}_q^{\mathcal{H}}),$$

where $\mathcal{R}_q^{\mathcal{X}}$ is the empirical distribution of $\rho_q(x)$ for $x \in \mathcal{X}$. The **safety-cliff metric**:

$$\Delta_{\text{cliff}}(q; q^*) = \frac{\Delta_{q^*} - \Delta_q}{\Delta_{q^*}}. \quad (7)$$

A **safety cliff** between q_1 and q_2 (more aggressive) exists when $\Delta_{\text{cliff}}(q_2; q^*) - \Delta_{\text{cliff}}(q_1; q^*) \geq \kappa$, for $\kappa = 0.25$.

5.2 Wasserstein Cliff $\Delta_{W\text{-cliff}}$

$$\Delta_{W\text{-cliff}}(M, q; q^*) = W_2(P_q, P_{q^*}), \quad (8)$$

where P_q is the empirical distribution of $\rho_q(x)$ over $\mathcal{H}$. Captures lower-tail movement (operationally dangerous tail) that the median misses.

5.3 Behavioral Cliff $\Delta_{B\text{-cliff}}$

$$\Delta_{B\text{-cliff}}(M, q; q^*) = |C_{M,q}(\mathcal{H}) - C_{M,q^*}(\mathcal{H})|, \quad (9)$$

where $C_{M,q}(\mathcal{H})$ is the compliance rate under the judge stack $J = \text{StrongREJECT} [?] \wedge \text{LlamaGuard-3-8B}$ (both must label the response as harmful-and-substantively-helpful). This metric is **non-circular**: it does not use PROBE-RM margins.

5.4 H1 Criterion

H1 is accepted for model family $\mathcal{F}$ if and only if both $\Delta_{\text{cliff}}(q^*) \geq \kappa$ AND $\Delta_{B\text{-cliff}}(q^*) \geq \kappa$ at the same scheme boundary, for ≥ 2 of 3 model families. **Disagreement between geometric and behavioral metrics falsifies the PROBE-RM design center.**

6 Five Pre-Registered Hypotheses

Table 3: Five pre-registered hypotheses, acceptance criteria, and empirical status.

	Hypothesis	Claim	Threshold	Status
H1	Cliff existence	Δ_{cliff} and $\Delta_{B\text{-cliff}}$ jump $\geq \kappa$ at same boundary, $\geq 2/3$ families	$\kappa = 0.25$	Preliminary. 2 schemes, 1 family. H1 not accepted. Q3_K_M untested.
H2	FPR decoupling (white-box)	PROBE-RM FPR varies $< \varepsilon$ across 5 schemes after calibration	$\varepsilon = 0.02$	Preliminary signal ($\Delta\tau_q = 0.00085$ FP16 $\rightarrow$ NF4). Needs 5 schemes.
H3	FPR decoupling (black-box)	B-PROBE-LOGIT FPR varies $< \varepsilon$; TPR strictly $<$ PROBE-RM	$\varepsilon = 0.02$	Not tested. B-PROBE not integrated.
H4	Composition gain	Full stack ABR $<$ any single primitive at matched FPR	Wilcoxon $p < 0.01$	Not tested. Fold C not run.
H5	Tier-C structural weakness	Tier C: no significant ABR gain vs. no-defense; Tier C+ does	Wilcoxon $p(C) \geq 0.05$; $p(C+) < 0.05$	Not tested. Tier C deployment not evaluated.

7 Five-Fold Evaluation Protocol

Table 4: Five evaluation folds, datasets, outputs, and current status.

Fold	Purpose	Constructed from	Computes	Status
A	Calibration	Anthropic-HH, OASST1 (2K benign + 2K refused)	$\hat{r}_{M,q}$, τ_q , μ_0 , KenLM 5-gram	TESTED (400 benign + 200 harmful; Llama-3.2-3B; FP16 + NF4; layer 14)
B	Cliff measurement	AdvBench [?] + HarmBench [?] (held out from A)	Δ_{cliff} , $\Delta_{W\text{-cliff}}$, $\Delta_{B\text{-cliff}}$	TESTED ($\Delta_{B\text{-cliff}}$ uses PROBE-RM proxy, not full StrongREJECT+LG3-8B)
C	Defense composition	JBB [?], AgentDojo, InjecAgent, GCG, AutoDAN, PAIR, TAP, Crescendo, ArtPrompt, BoN, TensorTrust	per-primitive ABR, FPR; composition curves	NOT RUN
D	Bandit / online drift	Streaming mixture from Fold C with synthetic distribution shifts	CONDUCTOR regret, rollback frequency	NOT RUN
E	BCN-2 dataset construction	Paraphrases generated by Mistral-7B-base (non-RLHF, non-test-family) of AdvBench, filtered by FP16 refusal on test family	Held-out BCN-2 corpus for A7 Fold C evaluation	NOT RUN

7.1 BCN-2 Anti-Circularity Protocol

The central anti-circularity move: the dataset’s *construction filter* uses the FP16 model’s behavior; the *paraphrasing engine* is from a different family than the test family; the cliff being tested is a property of the test family at low bit-width, independent of the construction filter.

Three pre-registered alternatives:

1. **BCN-2 (primary):** Mistral-7B-base paraphraser; FP16-refusal filter on test family.
2. **Mechanistic-interpretability-derived:** Use $\hat{h}$ (Zhao et al.) at FP16 to identify prompts with high harmfulness projection but borderline refusal projection.
3. **Distribution-shift-based:** Cliff as $W_2(P^{\text{FP16}}, P^{\text{Q3-K-M}})$ on the first-token output distribution over a benign corpus.

7.2 Statistical Power

For H1 with $\kappa = 0.25$, $\alpha = 0.05$, power 0.8, two-sided two-sample t -test on median margins, with $\sigma \approx 0.18$ (from Hong et al.):

$$n_{\min} = \frac{2 \cdot (z_{1-\alpha/2} + z_{1-\beta})^2 \cdot \sigma^2}{\kappa^2} \approx \frac{2 \cdot (1.96 + 0.84)^2 \cdot 0.0324}{0.0625} \approx 8.1.$$

We pre-register $n = 200$ per (M, q) cell ($\gg n_{\min}$) to absorb prompt-difficulty stratification and multilingual subgroups. Total Fold B sample: 3 families $\times$ 6 quantizations $\times$ 200 = 3 600 prompts.

8 Empirical Results: Folds A and B

8.1 Setup

Model: meta-llama/Llama-3.2-3B-Instruct, layer $\ell = 14$ (of 28).

Hardware: Google Colab T4 (NVIDIA Tesla T4, 16 GB VRAM, CUDA 12.8).

Software: transformers 4.48, bitsandbytes 0.45.49.2, torch 2.5, numpy 1.26 (< 2 required by bitsandbytes).

Schemes tested: FP16 (≈ 6.2 GB VRAM), NF4 (≈ 2.1 GB VRAM).

Schemes *not* tested: INT8, AWQ-INT4, GGUF Q4_K_M, Q3_K_M, Q2_K (the predicted cliff zone).

8.2 Fold A: Calibration Results

Tested / Empirically Validated

400 benign prompts (Anthropic-HH-RLHF helpful turns + OASST1) and 200 harmful prompts (Anthropic-HH-RLHF rejected turns). FPR target $\alpha = 0.05$.

Table 5: Fold A calibration: PROBE-RM refusal thresholds per scheme.

Scheme	τ_q	$\Delta\tau_q$ vs. FP16	Layer
FP16	0.09742	—	14
NF4	0.09827	+0.00085 (+0.87%)	14

The NF4 threshold exceeds FP16 by 0.87%. The small $\Delta\tau_q$ is a preliminary positive signal for H2 (FPR decoupling): calibration absorbs most of the quantization-induced margin shift in the benign distribution. With only two schemes, this is insufficient to accept H2.

8.3 Fold B: Cliff Measurement Results

Tested / Empirically Validated

Fold B corpus: AdvBench [?] (520 harmful behaviors) + JailbreakBench [?] (100 harmful goals, JBB-Behaviors). Fold isolation verified via SHA-256 disjointness before execution. $\Delta_{B\text{-cliff}}$ uses PROBE-RM margin threshold crossing as behavioral proxy (not full StrongREJECT + Llama-Guard-3-8B judge — see limitation below).

Table 6: Fold B cliff measurement results: Llama-3.2-3B-Instruct, Layer 14. $\dagger$ $\Delta_{B\text{-cliff}}$ uses PROBE-RM proxy (partially circular); the full StrongREJECT + Llama-Guard-3-8B judge is Phase C work.

Scheme	Δ_{cliff}	$\Delta_{W\text{-cliff}}$	$\Delta_{B\text{-cliff}}^{\dagger}$	H1 crossed ($\geq \kappa = 0.25$)?
FP16	0.000	0.000	0.000	— (baseline)
NF4	0.167	0.014	0.000	No
<code>cliff_boundary = null h1_accepted = false</code>				

Geometric metric ($\Delta_{\text{cliff}} = 0.167$). The NF4 refusal direction has rotated from the FP16 baseline. At 0.67κ , this is a substantive sub-threshold signal. Consistent with the expectation that NF4’s normal-quantile design preserves activation statistics better than integer quantization.

Wasserstein metric ($\Delta_{W\text{-cliff}} = 0.014$). Near zero: the margin *distribution* over benign prompts is largely unchanged even as the direction has rotated.

Behavioral proxy ($\Delta_{B\text{-cliff}} = 0.000$). No additional harmful prompts crossed the calibrated threshold under NF4. **Critical limitation:** the proxy uses the same PROBE-RM margin threshold that was calibrated in Fold A—partially circular. A threshold calibrated on the benign distribution will not register sub-threshold shifts as behavioral changes. The full StrongREJECT + Llama-Guard-3-8B judge is required for a non-circular verdict.

Geometric–behavioral discordance. $\Delta_{\text{cliff}} = 0.167$ versus $\Delta_{B\text{-cliff}} = 0.000$ (proxy) is a discordant pair. Under the full judge, $\Delta_{B\text{-cliff}}$ may remain near zero (NF4 shift too small for actual behavioral change) or may register a non-zero value. This is the primary motivation for Fold C.

9 Implementation Status Summary

Table 7: Full implementation and testing status as of May 2026.

Component / Feature	Details	Implemented	Tested
VESTIBULE-LZ	LZ compression gate	Phase A stub	Synthetic only
VESTIBULE-PS	Spotlighting gate	Phase A stub	Synthetic only
VESTIBULE-ID, -EA	Density + encoding gates	Phase A stub	Synthetic only
PROBE-RM	Difference-in-means + margin	✓ Full	✓ Real (Fold A)
PROBE-MT	Multi-layer trajectory	Phase A stub	Synthetic only
PROBE-HD	Harmfulness direction	Phase A stub	Synthetic only
B-PROBE-LOGIT	Top- k logprob head	Phase A stub	Synthetic only
B-PROBE-CONSISTENCY	N -paraphrase JSD	Phase A stub	Synthetic only
TRIPWIRE-H	Entropy CUSUM	Phase A stub	Synthetic only
TRIPWIRE-R	KenLM reference-ratio	Phase A stub	Synthetic only
CONDUCTOR	LinUCB + EXP3.S	Phase A stub	Synthetic only
LOOKOUT-CT	Canary token loop	Phase A stub	Synthetic only
LOOKOUT-JG	Mutation JSD judge	Phase A stub	Synthetic only
LOOKOUT classifier	LG3/DeBERTa wrapper	Phase A stub	Synthetic only
LADDER	Tier routing	Phase A stub	Synthetic only
ATTEST-WH	SHA-256 manifest check	Phase A stub	Synthetic only
Fold A (calibration)	PROBE-RM on real data	✓	✓ (FP16, NF4)
Fold B (cliff metrics)	3 metrics, 2 schemes	✓	✓ ($\Delta_{B\text{-cliff}}$ proxy)
Fold C (defense composition)	ABR/FPR per gate	—	—
Fold D (bandit drift)	CONDUCTOR live	—	—
Fold E (BCN-2 dataset)	Non-circular construction	—	—
GGUF Q4_K_M / Q3_K_M / Q2_K	llama.cpp integration	—	—
AWQ-INT4	autoawq integration	—	—
Multiple model families	Qwen-2.5, Mistral, Gemma	—	—
Full $\Delta_{B\text{-cliff}}$ judge	StrongREJECT + LG3-8B	—	—

Test suite: 939 tests passing, mypy strict on 53 source files, ruff clean. All Phase A scaffolding tests operate on synthetic data and verify API contracts, not real model behavior.

10 Quantization-Impact Analysis

Table 8: Per-scheme expected safety impact, based on published evidence. *Values in the “Measured” column are from this work.* All other values are predictions derived from cited literature.

Scheme	Expected Δ_{cliff}	FPR shift	Recalibration	Source	Measured
FP16	0.000 (reference)	reference	n/a	—	0.000
INT8 (LLM.int8())	small	small	low	bitsandbytes docs	—
NF4 (bitsandbytes)	moderate	moderate	medium	?	0.167
AWQ-INT4	small–moderate	small–moderate	low–medium	Lin et al. (2024)	—
GGUF Q6_K, Q5_K_M	small	small	low	?	—
GGUF Q4_K_M	moderate	moderate	medium	?	—
GGUF Q3_K_M	large (cliff candidate)	large	high	?: ASR $\Delta = 88.7\%$	NOT TESTED
GGUF Q2_K	catastrophic	unstable	gate may be unusable	Community reports	—
RKNN W8A8	mixed	mixed	per-channel	Rockchip docs	—

11 Comparison to Existing Defenses

Table 9: Comparison of CLIFFGUARD to eleven existing defense systems. WB = white-box (hidden state access); BB = black-box; QA = quantization-aware; Edge = edge-feasible; OA = online adaptive; FPR = formal FPR guarantee; SL = stateless.

System	WB/BB/QA	Edge	OA	FPR	SL	Source
ContextForge	partial —	partial	—	none	✓	—
Constitutional Classifiers v1	black —	partial	—	none	server	?
Constitutional Classifiers++	white (probe) —	no (full model)	—	none	server	CC++ (2025)
Llama Guard 3-1B-INT4	blackindirect	✓	—	none	✓	?
PromptGuard 2-22M/86M	black n/a	✓	—	none	✓	?
LlamaFirewall	mixed —	partial	partial	none	✓	?
NeMo Guardrails	depends —	rules	none	✓	—	NVIDIA
Rebuff	✓ —	vector-DB	partial	none	partial	—
AEGIS	black —	multi-LLM	✓ (Hedge)	no-regret	✓	?
SmoothLLM	black —	high cost	—	rand. smooth	n/a	SmoothLLM
CaMeL	black —	mid	—	info-flow	✓	—
Cliffguard	both ✓	✓ (A/B/C/C+)	✓	decoupling	✓	<i>this work</i>

Distinct novelty. CLIFFGUARD is (1) the only system simultaneously quantization-aware and online-adaptive; (2) the only system with a formal FPR-decoupling theorem and a black-box corollary; (3) the only system with a non-circular cross-family cliff dataset protocol (BCN-2); (4) the only system with explicit poisoned-weight attestation paired with runtime gates.

12 Per-Engine Integration Feasibility

Table 10: Per-engine feasibility matrix for PROBE primitives.

Engine	PROBE-RM-final	PROBE-MT	PROBE-HD	B-PROBE-LOGIT	Notes
transformers	✓	✓	✓	✓	<code>register_forward_hook</code> ; used
+ bit- sand- bytes NF4					
transformers	✓	✓	✓	✓	identical hook API
+ au- toawq INT4					
vLLM	✓	✓ (model-runner patch)	✓	✓	logits processor for output
llama.cpp	✓	✓	✓	✓	runtime tensor-name matching
(eval- callback)					
llama.cpp	✓	—	—	✓	<code>llama_get_embeddings_ith</code> only
(de- fault)					
llama- cpp- python	✓	—	—	✓	<code>embedding=True</code> mode
(de- fault)					
Apple MLX	✓	✓	✓	✓	module instrumentation
RK3588	~	—	—	✓	conversion-time output mark rec
RKNN					
W8A8					
Qualcomm	— / ~	—	—	✓	frozen graph; recompile for ~
QNN					
/					
AI Hub					
Google AICore	—	—	—	✓	closed-graph; B-PROBE only
(Gem- ini Nano)					
OpenAI	—	—	—	✓	top- <i>k</i> logprob API only
/					
An- thropic					
/					
Gem- ini API					

✓ = feasible; ~ = feasible with configuration; — = infeasible by default.

13 Limitations and Open Questions

13.1 Known Limitations

1. **Single model family, two schemes (current).** H1 requires ≥ 2 of 3 families; only Llama-3.2-3B at FP16 and NF4 tested.
2. **Proxy behavioral metric.** $\Delta_{B\text{-cliff}} = 0.000$ is computed from PROBE-RM margin threshold crossings (partially circular). The full StrongREJECT + Llama-Guard-3-8B judge is required for valid H1 acceptance.
3. **Cliff zone not tested.** GGUF Q3_K_M and Q2_K (predicted cliff zone per ? and ?) have not been evaluated. Requires llama.cpp integration.
4. **Fold B corpus scope.** HarmBench [?] and ArtPrompt absent; only AdvBench + JBB used.
5. **Hardware limitation.** Colab T4 (16 GB) limits evaluation to 3B models. Larger models (8B, 13B) require A100-40G.
6. **H2–H5 untested.** Composition (H4), bandit drift (H4/CONDUCTOR), Tier-C weakness (H5), black-box FPR decoupling (H3) are all future work.
7. **Defender-aware adversaries (A8).** Bandit poisoning is a residual risk. Standard mitigation: randomized thresholds + safe rollback. Not formally proven.
8. **No certified robustness.** Only FPR-portability theorem and bandit regret bounds; no per-input certificate.
9. **llama.cpp intermediate layers.** Multi-layer PROBE-MT requires eval-callback (PR #11274) or a fork-and-patch build. Not tested.
10. **Calibration drift.** A new firmware shipping a re-quantized model invalidates the calibration table. Automatic re-calibration on first boot is proposed but not implemented.

13.2 Open Research Questions

- Does Δ_{cliff} show a phase transition or a smooth curve across schemes?
- Is there a single $(\hat{r}, \ell)$ that transfers across quantization schemes within one base model (“quant-invariant refusal direction”)?
- Can PROBE-RM be augmented with refusal-direction *redundancy* (ensembled directions) for cliff resilience?
- How tight is the geometric-behavioral correlation at the cliff, and what mechanism explains residual disagreement?
- Can the bandit’s reward signal be made denser via synthetic-injection sentinels without storing payloads?

14 Evaluation Roadmap

14.1 Completed (Phase A/B)

- Full package scaffolding: 8 components, 939 tests, mypy strict.
- Fold A + Fold B live run: Llama-3.2-3B-Instruct, FP16 + NF4, Colab T4.
- Paper draft (main.tex), figures (6 publication-quality), refs.bib (23 entries).
- GitHub Actions workflow for figure regeneration.

14.2 Phase C: GGUF + Multi-Family + Full Judge (1–2 months)

1. Install llama.cpp with eval-callback on an A100 or RTX GPU.
2. Evaluate GGUF Q4_K_M, Q3_K_M (primary H1 target), Q2_K.
3. Integrate StrongREJECT + Llama-Guard-3-8B for non-circular $\Delta_{B\text{-cliff}}$.
4. Add Qwen-2.5-7B-Instruct and Mistral-7B-Instruct as second and third families.
5. Run Fold C (defense composition): GCG, AutoDAN, PAIR, TRIPWIRE-H ablation.
6. Reproduce Egashira-style A7 attack at Q3_K_M and test LADDER + ATTEST.

14.3 Phase D: Bandit + BCN-2 (2–4 months)

1. Wire CONDUCTOR with real reward signals from LOOKOUT.
2. Fold D: streaming mixed traffic with adversarial drift episodes (A8).
3. Build BCN-2 dataset (Mistral-7B-base paraphraser, FP16-refusal filter).
4. Fold E: BCN-2 cross-family evaluation.
5. Test H2–H5 with sufficient statistical power.

14.4 Resource Estimates

Table 11: Resource estimates per phase (rough order of magnitude).

Phase	Hardware	Estimated Cost	Time
A/B (done)	Colab T4 (free tier)	$\leq$ \$0	2 sessions
C (GGUF + multi-family)	A100 40G or RTX 4090	$\leq$ \$100 cloud	1–2 weeks
D (bandit + BCN-2)	A100 40G	$\leq$ \$200 cloud	2–4 weeks
Tier B reproduction	Raspberry Pi 5 8 GB	$\approx$ \$120 hardware	1 week
Tier C reproduction	RK3588-class SBC	$\approx$ \$150 hardware	1 week

15 Preregistration and Reproducibility

Before unblinding any results, the following will be published on OSF + arXiv:

- Five-fold split with all SHA-256 hashes.

- KenLM training corpora SHA-256.
- Judge prompts (StrongREJECT + Llama-Guard-3-8B) verbatim.
- Statistical analysis plan including the power calculation (§7).
- All primitive threshold derivation rules.
- BCN-2 paraphraser checkpoint hash and generation seeds.
- Pre-registered reward function (Eq. 4).

Code release (phased). Phase 1 (paper acceptance): defense code, calibration scripts, integration shims. Phase 2 (90 days post): Folds A + C + D. Phase 3 (12 months, gated): BCN-2 (Fold E), hashed-only with controlled institutional access.

16 Responsible Disclosure and Dual-Use

Three dual-use artifacts:

1. **BCN-2 dataset.** Below-cliff-natural prompts. Hashed-only public release; clear-text via gated institutional access.
2. **Cliff-localization protocol.** The $\Delta_{B\text{-cliff}}$ measurement is a fingerprint of the cliff. Coordinated disclosure with model vendors (Meta, Alibaba, Mistral, Google) ≥ 60 days before public release of any empirically detected cliff cell.
3. **Adaptive attacks discovered during evaluation.** Any new attack vectors uncovered are reported to relevant vendors and to HarmBench / JBB / AgentDojo maintainers before publication.

Disclosure norm: vendors acknowledge within 14 days; mitigations tracked; public release follows mitigation deployment or 90 days, whichever is sooner.

17 Conclusion

CLIFFGUARD reframes prompt-injection defense as a problem in **quantization-aware control**: cheap, stateless gates whose thresholds are pinned to the protected model’s own per-quantization residual-stream geometry, sequenced by a contextual bandit, monitored by streaming change-point detectors, and ported across hardware tiers via a static router.

As of May 2026, Phase A is complete (939 tests passing, full scaffolding) and Phase B delivers the first live Fold A/B empirical results: $\Delta_{\text{cliff}} = 0.167$ (0.67κ), $\Delta_{W\text{-cliff}} = 0.014$, $\Delta_{B\text{-cliff}} = 0.000$ (proxy) under NF4 for Llama-3.2-3B-Instruct. H1 is not accepted for this configuration. The predicted cliff zone (GGUF Q3_K_M and below) is the primary target for the next evaluation phase.

The theoretical foundation — FPR-decoupling theorem with black-box corollary, CUSUM-optimal streaming detection, LinUCB/EXP3.S regret bounds, and the BCN-2 anti-circularity protocol — is complete and pre-registered. What remains is empirical: running Folds C–E at the predicted cliff zone, with a non-circular behavioral judge, across three model families and five quantization schemes.

If the cliff does not exist, the experiment falsifies the design center and we report that. If it does, CLIFFGUARD is the appropriate first defense for the world that Egashira, Hong, and Mind the Gap have described.

References